Reading Material by Lectures, V1.0, TTK4147 2026

01 - Introduction:

Stallings 8th and 9th ed.:
1.1 - 1.2

Burns and Wellings:
1.1

02 - Hardware and Operating Systems:

Stallings 8th and 9th ed.:
1.3 - 1.7
2.1 - 2.4
3.1

03 - Multiprogramming and Memory Management

Stallings 8th and 9th ed.:
3.1 - 3.4
3.7
4.1 - 4.3
4.9

Stallings 8th ed.:
5.1 - 5.3
5.5
5.7

Stallings 9th ed.:
5.2 – 5.4
5.6
5.8

Stallings 8th and 9th ed.:
6.1 - 6.4
6.12
7.1 - 7.5
8.1
8.7

04 - Scheduling

Stallings 8th and 9th ed.:
9.1-9.2
9.4

Burns and Wellings:
11.1
11.2
11.3
11.4
11.5

05 – Deadlocks, Mixed Criticality and FreeRTOS

Stallings 8th and 9th ed.:
6.1-6.4
6.6
6.12

Burns and Wellings:
1.1 - 1.3
11.1 - 11.5

(The 13 following links are checked 2026-08-13.)

1. **FreeRTOS - a brief overview**
   https://www.cs.unc.edu/~anderson/teach/comp790/powerpoints/freertos-proj.pdf
2. Tasks and Co-routines
   http://www.freertos.org/taskandcr.html
3. FreeRTOS task states, behavior, implementation and priorities explained
   https://www.freertos.org/Documentation/02-Kernel/02-Kernel-features/01-Tasks-and-co-routines/01-Tasks-overview
4. FreeRTOS task states and state transitions described
   http://www.freertos.org/RTOS-task-states.html
5. RTOS task priorities in FreeRTOS for pre-emptive and co-operative real time operation
   http://www.freertos.org/RTOS-task-priority.html
6. Writing RTOS tasks in FreeRTOS - implementing tasks as forever loops
   http://www.freertos.org/implementing-a-FreeRTOS-task.html
7. RTOS idle task and the FreeRTOS idle task hook function
   http://www.freertos.org/RTOS-idle-task.html
8. FreeRTOS Inter-task communication
   https://www.freertos.org/Documentation/02-Kernel/02-Kernel-features/03-Direct-to-task-notifications/06-Inter-Task-Communication
9. FreeRTOS task notifications, Inter-task communication and synchronisation
   http://www.freertos.org/RTOS-task-notifications.html
10. FreeRTOS task notifications, Used As Light Weight Binary Semaphores
    http://www.freertos.org/RTOS_Task_Notification_As_Binary_Semaphore.html
11. FreeRTOS task notifications, Used As Light Weight Counting Semaphores
    http://www.freertos.org/RTOS_Task_Notification_As_Counting_Semaphore.html
12. FreeRTOS task notifications, used as light weight event group
    http://www.freertos.org/RTOS_Task_Notification_As_Event_Group.html
13. FreeRTOS task notifications, Used As Light Weight Mailbox
    http://www.freertos.org/RTOS_Task_Notification_As_Mailbox.html

06 - Priority Inversion

Burns and Wellings:
11.8 - 11.9

07 - Linux

Stallings 8th and 9th ed.:
2.10
4.6
6.8
8.4
10.3

How fast is fast enough.pdf
Xenomai 3 an overview of the real-time framework for linux.pdf

Provided on Canvas

08 - More Scheduling

Stallings 8th and 9th ed.:
10.1 - 10.2
10.7

Burns and Wellings:
11.6 - 11.7
11.10 - 11.11

09 - QNX

QNX Neutrino RTOS System Architecture.pdf:
chapter 1, 2, 3, 4, 6, 8, 13

(The following link is checked 2024-08-20)

Transparent Distributed Processing Using Qnet
http://www.qnx.com/developers/docs/6.3.2/neutrino/prog/qnet.html

10 - Distributed Systems – Architecture

MSIT_116D_Distributed_Systems

Chapter 1-3
Provided on Canvas

11 – Guest lecture

Guest lecture

Only slides

12 - Analysis of Physical systems

Only slides